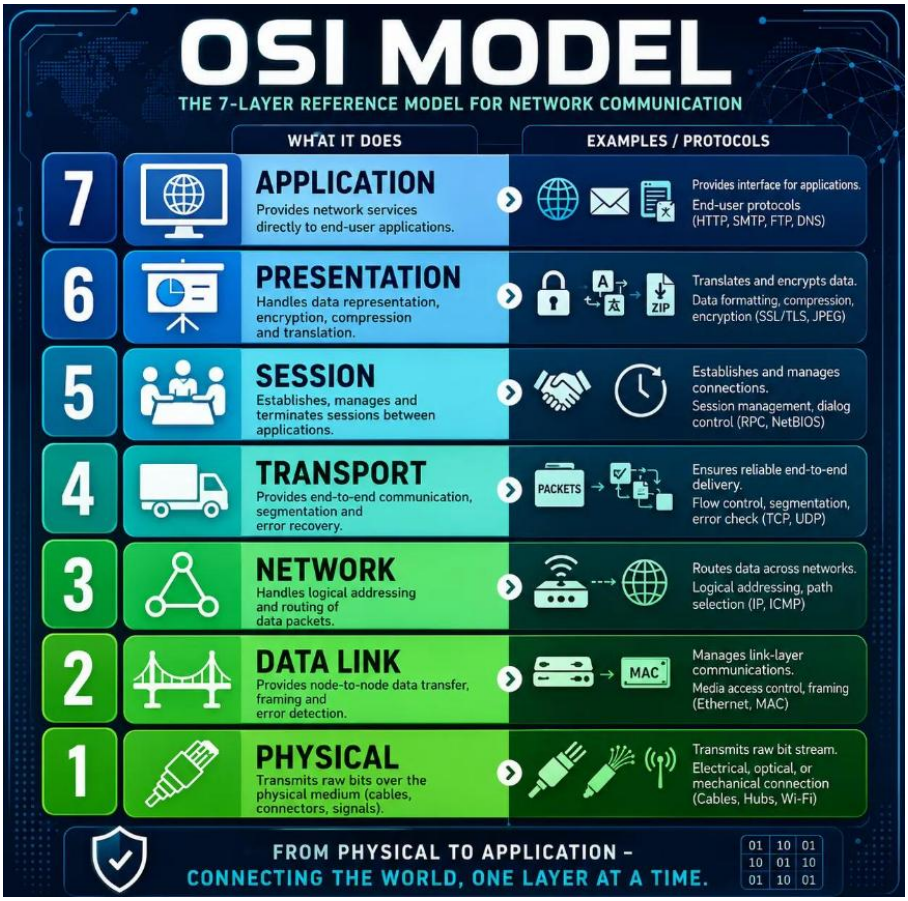
OSI Model & Network Protocols

Task – 1 : Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

Solution :



Task – 2 : Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

Solution : Different applications use networking protocols to communicate with their servers, exchange information, and transfer data.

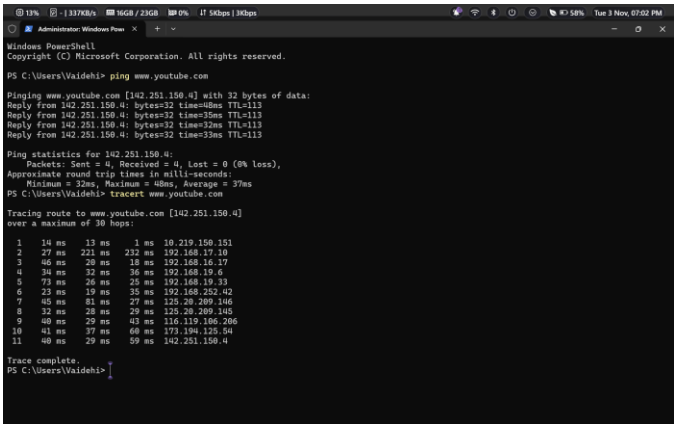
Application	**Protocol 1**	**Protocol 2**	**Purpose**
WhatsApp	HTTPS	TCP	Secure communication with WhatsApp services and reliable data transfer.
Instagram	HTTPS	TCP	Loads posts, reels, images, and account information securely.
Zomato	HTTPS	DNS	Secure restaurant searches and domain name resolution.
Flipkart	HTTPS	TCP	Secure shopping, product browsing, and data transmission.
Paytm	HTTPS	TLS	Protects sensitive information during online transactions.

Explanation:

1. HTTPS: Secures communication between applications and servers.
2. TCP: Provides reliable delivery of data between devices.
3. DNS: Converts website domain names into IP addresses.
4. TLS: Encrypts communication to protect sensitive information.

Task – 3 : Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

Solution :



OSI Layers Involved in Ping and Tracert

OSI Layer	**Ping**	**Tracert**
Application (7)	Command is entered in CMD. DNS may resolve the domain name.	Command is entered in CMD. DNS may resolve the domain name.
Transport (4)	Does not use TCP or UDP.	Windows tracert uses ICMP probes rather than TCP or UDP.
Network (3)	Uses IP and ICMP to send and receive packets.	Uses IP and TTL values to identify network hops.
Data Link (2)	Uses Ethernet or Wi-Fi frames to transmit packets locally.	Uses Ethernet or Wi-Fi frames to transmit packets locally.
Physical (1)	Transmits data through cables or wireless signals.	Transmits data through cables or wireless signals.

Task – 4 : Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step. Hint : Start from Application layer (your message in WhatsApp) down to Physical layer (actual data sent over WiFi/mobile network), then back up to Application layer on your friend's phone.

Solution : **Step-by-step communication**

Sender's Device (My Smartphone)

1. Application Layer (Layer 7): I type the message "Hello, how are you?" in WhatsApp and press the Send button.

2. Presentation Layer (Layer 6): The message is protected using encryption. WhatsApp uses end-to-end encryption to protect personal messages.

3. Session Layer (Layer 5): Communication between the devices and WhatsApp's services is managed so that messages can be exchanged.

4. Transport Layer (Layer 4): Data is transmitted using a transport protocol such as TCP, which provides reliable delivery, or another supported transport mechanism.

5. Network Layer (Layer 3): IP addresses are used to route packets through different networks towards WhatsApp's servers.

6. Data Link Layer (Layer 2): Data is converted into frames and transmitted through the local Wi-Fi or mobile network using hardware addressing.

7. Physical Layer (Layer 1): The data is transmitted as electrical, radio, or optical signals through the network.

Receiver's Device (Friend's Smartphone)

When the message reaches my friend's smartphone, the process works in reverse:

1. Physical Layer: Receives the transmitted signals.
2. Data Link Layer: Reconstructs the frames and checks their delivery.
3. Network Layer: Processes IP packets and delivers them to the appropriate destination.
4. Transport Layer: Reassembles the data and handles transport-level delivery.
5. Session Layer: Manages the communication context.
6. Presentation Layer: Handles decryption so the message can be read.
7. Application Layer: WhatsApp displays "Hello, how are you?" on my friend's screen.